

Bloomberg Primer - Here's How Biocomputing Works And Matters For AI

video from 0min to 8min33 :<https://www.youtube.com/watch?v=txtDpCLHUKU>

Key words and sentences :

- Human brain cells on a silicon chip can play Pong.
- Lab researchers trained a small cluster of neurons to play a video game.
- Merging biology with computing could revolutionize AI.
- Supercomputers and AI still rely on traditional silicon chips.
- Moore's Law doubled transistor density every two years for decades.
- Transistors are now one atom thick, reaching physical limits.
- Biological brains learn environments with far less data than AI.
- Reducing data and energy needs is the ultimate goal for AI.
- Running current AI models requires a massive amount of energy.
- A supercomputer uses up to 40 megawatts; a human brain uses only 20 watts.
- In living cells, hardware and software are dynamically connected.
- Silicon chips process tasks sequentially, while brains process information in parallel.
- Biological computing is extremely promising, but it is still in its infancy.

Abstract :

Lab researchers have demonstrated that a small cluster of human brain cells on a silicon chip can play Pong, proving that merging biology with computing could completely revolutionize artificial intelligence. Today, supercomputers and AI still rely on traditional silicon chips, but Moore's Law is reaching its physical limits as transistors become just one atom thick. At the same time, running modern AI models consumes massive amounts of energy—a supercomputer requires up to 40 megawatts, whereas a biological brain runs on just 20 watts and learns complex environments with far less data. Unlike silicon chips that process tasks sequentially, living cells

dynamically intertwine hardware and software to process information fluidly, making biological computing a game-changing alternative even though the technology is still in its infancy.

Opinion :

I chose this topic because AI is playing an increasingly prominent role in our daily lives, and I wanted to explore its current limitations and how we might overcome them. Up until now, computer science has tried to emulate the human brain, but a key question arises: why not use its raw material directly? The main advantage lies in efficiency—biological systems require far less data, learn faster, and consume significantly less energy, which is a major asset given today's climate challenges. However, this raises critical ethical questions: does a cluster of brain cells grown outside a body still possess a form of consciousness, and could this lead to a new form of exploitation?